

1559. Detect Cycles in 2D Grid

Solved ●

Medium Topics Hint

Given a 2D array of characters `grid` of size `m x n`, you need to find if there exists any cycle consisting of the **same value** in `grid`.

A cycle is a path of **length 4 or more** in the grid that starts and ends at the same cell. From a given cell, you can move to one of the cells adjacent to it - in one of the four directions (up, down, left, or right), if it has the **same value** of the current cell.

Also, you cannot move to the cell that you visited in your last move. For example, the cycle `(1, 1) -> (1, 2) -> (1, 1)` is invalid because from `(1, 2)` we visited `(1, 1)` which was the last visited cell.

Return `true` if any cycle of the same value exists in `grid`, otherwise, return `false`.

Example 1:

a	a	a	a
a	b	b	a
a	b	b	a
a	a	a	a

Input: `grid = [[["a","a","a","a"],["a","b","b","a"],["a","b","b","a"],["a","a","a","a"]]`

Output: `true`

Explanation: There are two valid cycles shown in different colors in the image below:

a	a	a	a
a	b	b	a
a	b	b	a
a	a	a	a

Example 2:

c	c	c	a
c	d	c	c
c	c	e	c
f	c	c	c

Input: `grid = [[["c","c","c","a"],["c","d","c","c"],["c","c","e","c"],["f","c","c","c"]]`

Output: `true`

Explanation: There is only one valid cycle highlighted in the image below:

c	c	c	a
c	d	c	c
c	c	e	c
f	c	c	c

Example 3:

a	b	b
b	z	b
b	b	a

Input: `grid = [[["a","b","b"],["b","z","b"],["b","b","a"]]`

Output: `false`

Constraints:

- `m == grid.length`
- `n == grid[i].length`

- `1 <= m, n <= 500`
- `grid` consists only of lowercase English letters.

Seen this question in a real interview before? 1/6

Yes No

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Topics	▼
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